

Raspberry Pi 4 Model B : PinOut, Feat Specifications, Interfacing, Difference Applications

The **Raspberry Pi** is a credit-card-sized, versatile computer used in various education, basic computing, and more advanced **Raspberry Pi projects**. These sized, and popular boards that interact with the physical world through their General Purpose Output (GPIO) pins. Raspberry Pi boards are available in different models, including the flagship series, which includes Model A+, Raspberry Pi 4 Model B, the nano series, compact & low-power, the Compute Module series, the Pico series, and the Raspberry Pi 400. Among them, the fourth-generation Raspberry Pi, like the Raspberry Pi 4, is the most popular single-board computers. This article elaborates on the **Raspberry Pi 4 Model B**, its features, and applications.

What is the Raspberry Pi 4 Model B?

The Raspberry Pi 4 Model B is a small, low-cost single-board computer developed by the Raspberry Pi Foundation. It is the fourth-generation of the Raspberry Pi series and is designed to be a versatile, credit-card-sized computer that can be used for a wide range of applications, from education to professional development.

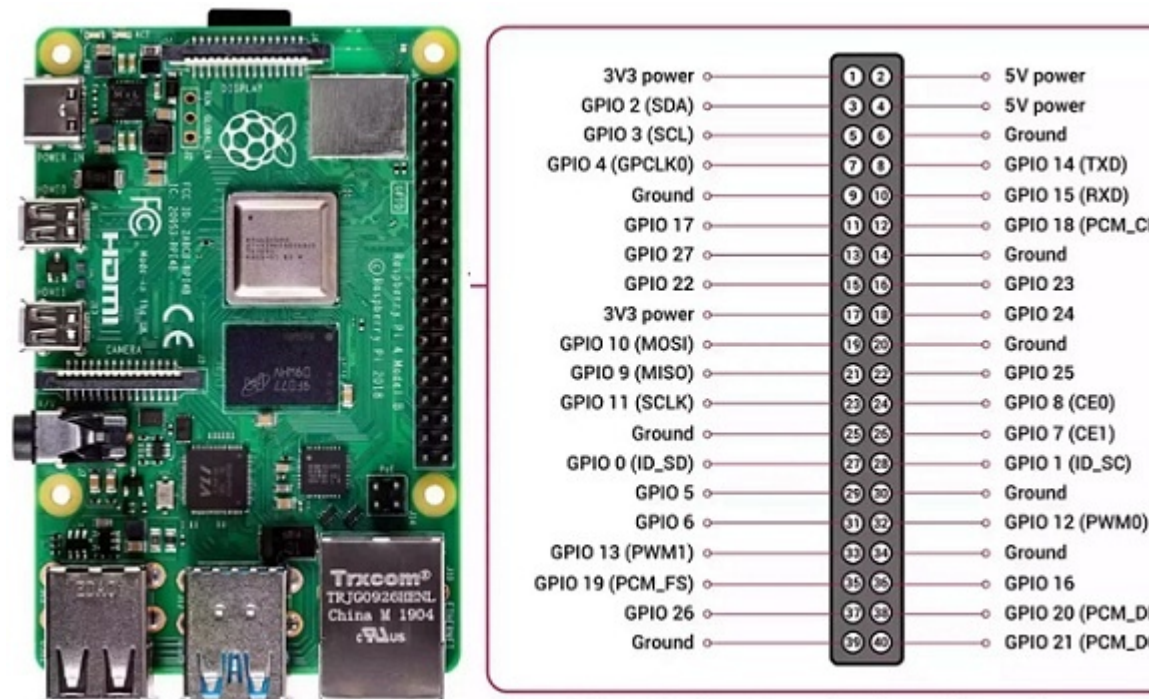
Pi Foundation. This board is a versatile platform for creating, learning & experimenting widely in lightweight desktop PC, DIY projects, education, **home automation**, etc. The board has a 64-bit quad-core processor, up to 4K resolution dual-display, networking like dual-band Wi-Fi, Bluetooth 5.0 & Gigabit Ethernet, and several USB

How Raspberry Pi 4 Model B Work?

The Raspberry Pi 4 Model B basically works like a small and low-power computer operating systems, like Raspberry Pi **OS**, to perform many tasks. This can be random access memory, a central processing unit, a graphics processing unit, and ports, all on a single board.

Raspberry Pi 4 Model B Pin Diagram

The Raspberry Pi 4 Model B pin diagram is shown below. This board includes 40 pins. The pins incorporated into this module is discussed below.



Raspberry Pi 4 Model B Pin Diagram

Power & Ground Pins

This Raspberry Pi board includes power pins and GND pins like 5V, 3V3 (3.3V), you have a humidity sensor or PIR sensor, then these power pins are used to power

GPIO Pins

These are general-purpose I/O pins, used to connect with external devices. arranged to either general-purpose i/p or output pins. External labels from GF available with the BCM (Broadcom) naming convention. This convention is helpful going to program through Python libraries. Internal labels from 1 to 40 project convention. So this convention is helpful whenever Broadcom is not supported. S programming libraries.

SPI Pins

This module features SPI **communication protocols** that facilitate master-slave communication. pins play a crucial role in demonstrating how the controller interacts with peripheral sensors and shift registers. Specifically, the MISO (master input slave output) and output slave input) pins are essential for **SPI communication**.

Data synchronization occurs through the SCLK signal at pin GPIO11 from the Raspberry Pi module sends data to the SPI component via the MOSI pin. If the component respond, it transmits data back to our module using the MISO pin.

I2C Pins

This Raspberry Pi module is integrated with the I2C communication protocol, which uses two pins: SCL and SDL. The serial clock line or SCL, ensures the data transfer synchronous bus, whereas the serial data pin or SDL carries the data. So this communication involves master-slave roles between the devices.

UART Pins

This Raspberry Pi board includes a UART serial communication protocol where it uses two pins: Tx & Rx. The transmission (Tx) pin is used for the serial data transmission, and the receiving (Rx) pin guarantees the reception of serial data.

Features & Specifications:

The features and specifications of the Raspberry Pi 4 Model B include the following:

- Raspberry Pi 4 Model B is a 64-bit high-performance quad-core processor.
- This module needs a 5.1-volt, 3-amp power supply using GPIO or USB-C.
- This board includes 40 pins.
- It is available with BCM2711 Broadcom, a Quad-core ARMv8 Cortex-A72 with 1.5GHz processor.
- It supports 5.0 GHz & 4 GHz IEEE 802.11ac wireless and 5.0 GHz Bluetooth.

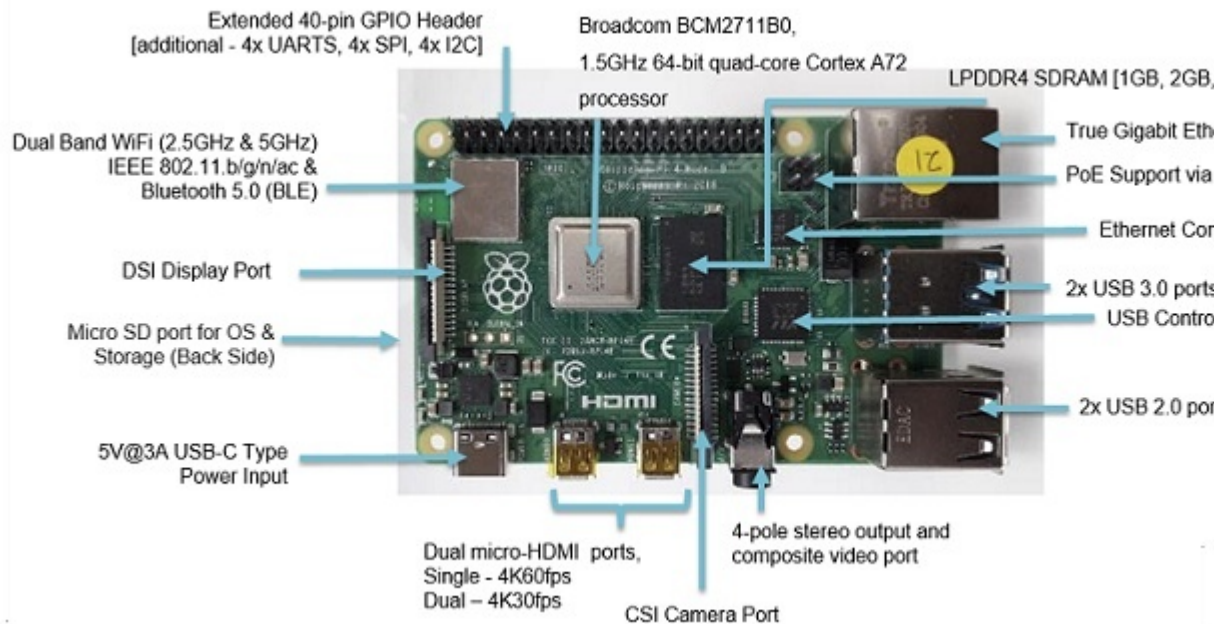
- This module carries Gigabit Ethernet to connect computers within a physical space.
- RAM is accessible with different types of memory.
- It includes an MIPI DSI 2-lane display port & MIPI CSI 2-lane camera port.
- This module includes one USB 3.0 port, USB-C power port & USB 2.0 port.
- It includes two micro-HDMI ports that support a 4K display.
- This module has a 4-pole stereo audio & composite video port.
- It includes H.264 & H.265 video coding.
- It includes a micro-SD card slot to load the OS & data storage.
- Operating temperature ranges from 0 to 50 degrees C ambient.
- It includes two HDMI micro ports, which support up to 4K 60Hz video resolution.
- It has 2-way MIPI DSI/CSI ports mainly for the camera & display.

How to Set Up Raspberry Pi 4 Model B?

For the Raspberry Pi 4 Model B setup, a microSD card is required with the OS, keyboard, a mouse & a monitor. Initially, flash the Raspberry Pi operating system card with the Raspberry Pi Imager. After that, the microSD card must be inserted connected to peripherals like a keyboard, monitor & mouse & given a power. Raspberry Pi must be configured through SSH or a graphical interface for remote access.

Raspberry Pi 4 Model B Components

Raspberry Pi 4 Model B includes different components, which are explained below



Raspberry Pi 4 Model B Components

BCM2711 SOC

The Broadcom BCM2711 SOC is the heart of the Raspberry Pi 4, which provides new features. This is a SoC (System on Chip) that provides power to the Raspk significant upgrade to other Raspberry Pi models with a 64-bit quad-core ARM Co

LPDDR4 RAM

LPDDR4 is a double-data-rate and low-power RAM technology, specially design smartphones. It offers power efficiency and performance balance, which is s powered applications. This RAM provides improved energy efficiency and hi compared to its predecessor.

MXL7704

The MXL7704 is a PMIC (power management integrated circuit) utilized in the Ra B. This IC provides all the required power rails for the board to simplify the des potentially decrease costs. The MXL7704 IC handles voltage regulation & distri the board gets accurate power for its variety of components.

BCM54213PE

The Gigabit Ethernet transceiver chip of the Raspberry Pi 4 Model B. like BCM5

chip. The main function of this IC is to handle all the physical layer functions, mainly 10BASE-T, and 1000BASE-T Ethernet connections above standard Category 5 UTP

This IC replaces the shared USB & Ethernet setup to provide true Gigabit Ethernet works as the interface between the Raspberry Pi's processor & the physical changing digital signals into the required format for transmission over the network.

VL805

The VL805 in the Raspberry Pi 4 Model B is a USB 3.0 host controller IC that is handling the USB ports of the board. It especially provides the interface for the two USB 3.0 ports by allowing them to communicate with the PCI Express platform bus. This chip allows 5 Gbps super-speed data transfer for USB 3.0 devices.

CSI Camera Port

A Camera Serial Interface (CSI) port is a specialized connector that allows a camera to capture and send data to the main processor. So it is a low-power and high-speed interface designed for image sensors and camera modules. Here, the CSI protocol is frequently used as MIPI CSI-2, which allows efficient image & video data transmission at higher data rates.

DSI Port

The 15-pin MIPI Display Serial Interface port in the Raspberry Pi 4 Model B is used to connect compatible displays. So this port allows for low-power and high-speed data transfer between the Raspberry Pi & a display with a 15-pin ribbon cable. This DSI interface is used to connect touch screens and also other displays in embedded systems.

Dual Mode HDMI Port

The Raspberry Pi 4 Model B includes a dual-mode HDMI port that supports up to two simultaneous dual display output. This board allows users to connect two separate monitor ports to HDMI cables or a blend of micro HDMI ports to HDMI adapters & normal HDMI cables. The Raspberry Pi 4 board can also have a MIPI DSI connector, used for a third display.

display setups.

Dual-band Wi-Fi

Raspberry Pi 4 Model B includes dual-band 802.11ac wireless LAN, which support 2.4 GHz & 5 GHz frequencies. So this allows for quicker networking with low interference as compared to models that support 2.4 GHz only.

Ethernet Controller

An Ethernet controller is a component that allows a device or computer to connect to a network to communicate with other devices on the network. This controller works as a bridge between the internal circuitry & the physical Ethernet cable of the device. Basically, it handles the low-level tasks of data transmitting and receiving over the network, like data packet management & error detection handling.

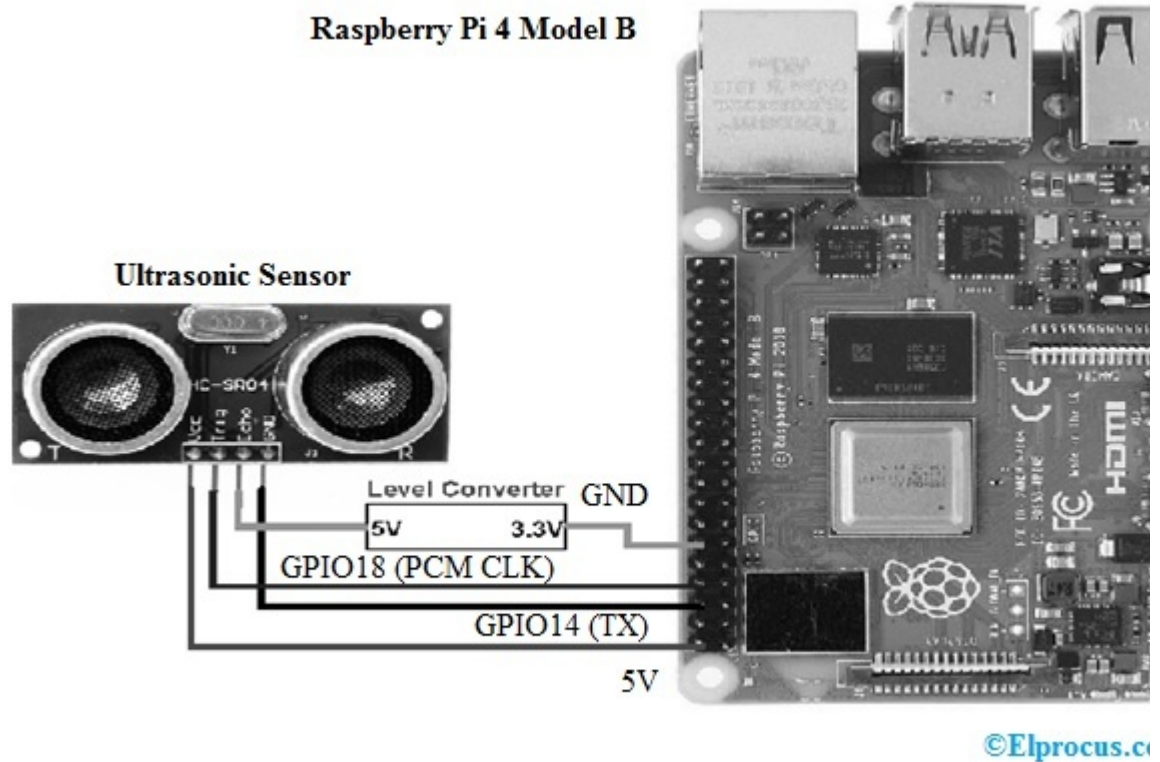
Raspberry Pi 4 Model B Interfacing with Ultrasonic Sensor

The Raspberry Pi 4 Model B interfacing with an ultrasonic sensor with Python code is given below. Generally, an ultrasonic sensor is used to measure distance. These sensors are used in obstacle avoidance robots & DIY radar projects.

Components & Connections:

The required components to make Raspberry Pi 4 model B with an ultrasonic sensor are:

- Raspberry Pi 4, ultrasonic sensor, voltage level converter, and 5V 3A power adapter.
- Connect the VCC pin of the ultrasonic sensor to the 5V pin of the Raspberry Pi 4.
- Connect the Trig pin of the ultrasonic sensor to the GPIO14 (TX) pin of the Raspberry Pi 4.
- Connect the Echo pin of the ultrasonic sensor to the GPIO18 (PCM-CLK) pin of the Raspberry Pi 4 Model B.
- Connect the GND pin of the ultrasonic sensor to the GND pin of the Raspberry Pi 4.



Raspberry Pi 4 Model B Interfacing with Ultrasonic Sensor

Code

The required Python code for the ultrasonic sensor is given below.

```
import RPi.GPIO as GPIO / This line begins with introducing modules to work with
import time
GPIO.setmode(GPIO.BCM) / Defines the numbering system used to function on the
GPIO_TRIG = 11 / Initialize the PINs where the ultrasonic sensor is connected/.
GPIO_ECHO = 18
GPIO.setup(GPIO_TRIG, GPIO.OUT) /Configure the pins of Raspberry Pi as input
pins/.
GPIO.setup(GPIO_ECHO, GPIO.IN)
GPIO.output(GPIO_TRIG, GPIO.LOW)
Time.sleep(2) / make the GPIO pin low for two milliseconds/
GPIO.output(GPIO_TRIG, GPIO.HIGH) / Introducing the delay to make GPIO_TRIG
Time.sleep(0.00001)
GPIO.output(GPIO_TRIG, GPIO.LOW)
while GPIO.input(GPIO_ECHO)==0:
```

```
start_time = time.time()
while GPIO.input(GPIO_ECHO)==1:
    Bounce_back_time = time.time()
    pulse_duration = Bounce_back_time - start_time
    distance = round(pulse_duration * 17150, 2)
    print (f"Distance: {distance} cm") /measure the time taken through the pulse
    receiver once hitting an object/
GPIO.cleanup()
```

Working

Ultrasonic sensor function in the above interfacing is similar to a radar. A transmitter in radar produces radio waves, an electromagnetic signal that travels and returns whenever it strikes an object in its path. After that, the distance is calculated using the formula as given below.

$$\text{Distance} = \text{Time} \times \text{Speed}$$

This sensor includes a transmitter and a receiver, where the transmitter is used to send sound waves and the receiver receives reflected sound signals.

Raspberry Pi 4 Model B Vs Raspberry Pi 5 4 GB

The difference between Raspberry Pi 4 Model B and Raspberry Pi 5 4 GB includes

Raspberry Pi 4B	Raspberry Pi 5
This board includes the Broadcom BCM2711 processor, including 4x Cortex-A72 CPU @ 1.8 GHz, whereas VideoCore VI GPU @ 500 MHz.	It includes the Broadcom BCM2712 processor, including 4x Cortex-A76 CPU @ 2.0 GHz, whereas VideoCore VII GPU @ 800 MHz.
Its display output is 4K 60Hz or two 4K 30Hz displays through miniHDMI ports.	It displays two 4K 60Hz display outputs through miniHDMI ports.

It includes 4GB LPDDR4 RAM.	This board uses faster 4GB & 8GB options with a 3631 MiB/sec transfer rate.
It depends on the major processor for I/O tasks with slower USB 3.0 speeds, a single CSI & DSI port.	This board has a committed RF chip to handle Ethernet & USB data transfer rates.
It lacks most of the features compared to the Raspberry Pi 5 4 GB.	It includes some features like: a real-time clock (RTC) module & an optional active thermal management.
It includes a MicroSD card slot with 52 Mbps.	It includes a microSD card slot
This board includes 5V 3A minimum power through the USB-C port.	This board includes 5V 5V 5A power through the USB-C port.

Advantages

The advantages of the Raspberry Pi 4 Model B include the following.

- Its increased processing power provides a significant performance boost for multitasking & quicker demanding task execution.
- This board has dual-band 2.4/5.0 GHz Wi-Fi & Gigabit Ethernet for more network connections.
- It is available in different RAM options.
- It supports dual micro-HDMI ports, which allows users to connect displays simultaneously.
- Its faster processor provides a significant performance.
- This board has multimedia performance.
- The upgraded GPU of this board allows for level 4K video playback to make it a home theater setups & media centers.
- The Pi 4 board has less power consumption.
- It is used in a wide range of applications.

Disadvantages

The **disadvantages of the Raspberry Pi 4 Model B** include the following.

- This Modelboard does not have any built-in storage. So it needs a microSD storage.
- It does not have a dedicated GPU (graphics processing unit), which restricts it to handle graphically demanding tasks.
- This board can overheat, particularly when performing severe tasks.
- Processing power is limited.
- Its performance is frequently limited by the microSD card speed utilized for storage.
- This board utilizes a USB-C port for power and needs a power adapter that provides a minimum power at 5V.

Applications

The applications of the Raspberry Pi 4 Model B include the following.

- **Home automation projects** utilize the Raspberry Pi 4 Model B to control appliances, security systems through IoT technology.
- Media centers stream movies, TV shows, and music using Kodi or Plex software on the board.
- RetroPie emulators allow users to enjoy classic games, while the board also serves as a storage solution or website host.
- You can even use it as a desktop computer for word processing and web browsing. For learning and development, this board teaches electronics, programming, and computer science.
- It creates a storage solution mainly for your files.
- It controls robots for hobbyists or educational purposes.
- This board is useful for experimenting with new software and hardware ideas.

The **Raspberry Pi 4 Model B board** is a significant upgrade to other models. With substantial developments to process power, memory, connectivity & multimedia capabilities, to achieve desktop-level performance, the Raspberry Pi 4 Model B supports a wide range of applications from hobbyist projects to advanced automation solutions.. Here is a question for you: What are the applications of Raspberry Pi?

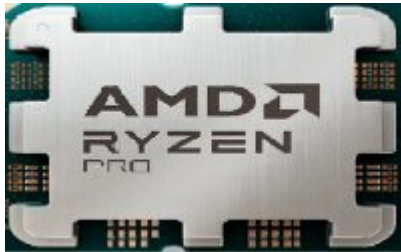
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